Candidate Number……..

THE PRESIDENT'S OFFICE – RALG

TAHOSSA ROMBO DISTRICT EXAMINATION SYNDICATE

FORM FOUR MOCK EXAMINATION

032/1 **CHEMISTRY 1**

TIME: 3:00 Hours **August, 2023**

Instructions

1. This paper consists of sections **A**, **B** and **C** with total **eleven (11)** questions.
2. Answer all questions in section A and B and **two (2)** questions from section C.
3. Section A carry **Sixteen (16)** marks, Section B carry **fifty-four (54)** marks and Section C carry **thirty (30)** marks
4. Cellular phones and any unauthorized materials are not allowed in examination room.
5. Write your **examination number** on every page of your answer sheet(s)
6. The following constants may be used: -
   Atomic mass: H=1, C=12, N=14, O=16, Na=23, S=32, Mg=24, Cl=35.5, Cu=64
   Avogadro's number = $6.02 \times 10^{23,}$ GMV at S.T.P = $22.4dm^3$, 1Faraday = 96,500coulombs.
   Standard pressure = 760mm Hg, Standard temperature = 273K, 1 litre = $1dm^3 = 1000cm^3$.

FOR ASSESSOR'S ONLY		
QUESTION NUMBER	**SCORE**	**MARKER'S INITIALS**
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
TOTAL		
CHEKER'S INITIALS		

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Candidate Number……...

SECTION: A (16marks)

1. For each of the items (i) – (x), choose the correct answer from among the given alternatives and write its letter besides the item number in the answer sheet provided

(i) Manufacturing of fertilizer is one of importance of chemistry from the following field

A. Agriculture
B. Transport
C. Medicine
D. Cosmetic industry
E. Telecommunication

(ii) The atoms of different chemical elements that have the same mass number but different atomic number are called ……

A. Isotopes
B. Isotopy
C. Isobar
D. Isomers
E. Allotropes

(iii) The one in which an atom or group of atoms is replaced by another atom or group of atoms in organic molecule or compound

A. Addition reaction
B. Elimination reaction
C. Combustion reaction
D. Cracking reaction
E. Substitution reaction

(iv) …… added to sodium chloride to lower melting point of sodium chloride from 7740°C to about 6000°C during extraction of sodium

A. Calcium chloride
B. Calcium oxide
C. Lime stone
D. Coke
E. Hematit

(v) It includes a statement that either prove or disprove the hypothesis

A. Data collection
B. Experimentation
C. Hypothesis
D. Drawing conclusion
E. Data interpretation

(vi) Water change to vapor absorb energy (heat) from surroundings and final cause cooling effects

A. Water cycle
B. Refinery
C. Steam engine
D. Refrigeration
E. Drying materials

(vii) The chemical substance which is neutral to litmus paper and turn white anhydrous copper sulphate to blue solution

A. Vinegar
B. Water
C. Oxygen
D. Carbon dioxide
E. Sulphur dioxide

(viii) The following is a metal compound that can be used to remove hardness of water

A. Calcium carbonate
B. Calcium hydroxide
C. Calcium oxide
D. Calcium chloride
E. Ammonium solution

(ix) Silicon and Germanium are good examples of ……..

A. Metal
B. Metalloids
C. Non – metal
D. Noble gas
E. Halogens

(x) The tendency of the atom to attract an electrons toward itself from an atom or ion

A. Atomic radius
B. Melting point
C. Ionization energy
D. Electro positivity
E. Electro negativity

2. Match the items in LIST A with those corresponding items in LIST B by writing the letter of correct response besides the item number in the answer sheet provided

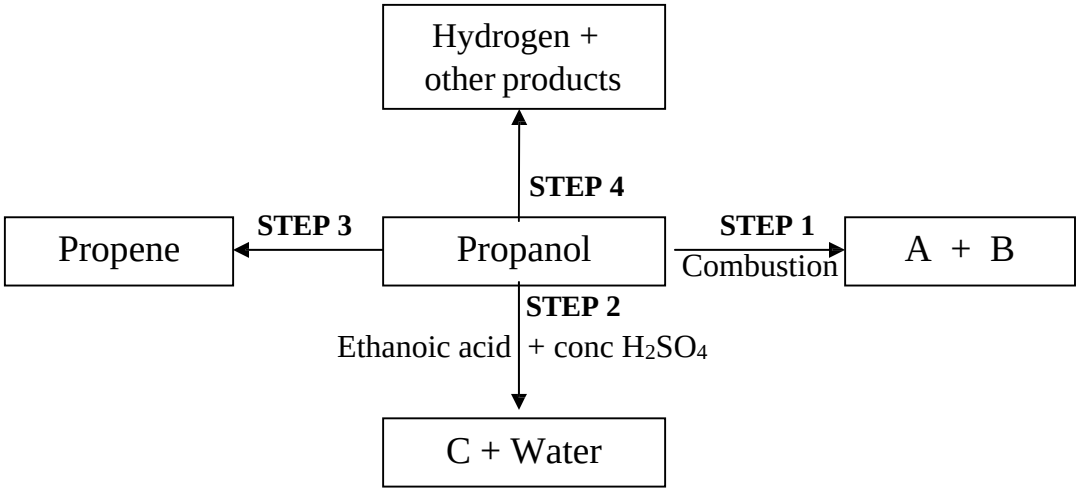
LIST A	**LIST B**
(i) Frasch process	A. Sulfur
(ii) Contact process	B. Hydrochloric acid
(iii) Ostwald process	C. Ammonia
(iv) Hydrolysis of carbon dioxide	D. Carbonic acid
(v) Naturally occur in stomach in gastric juice	E. Nitric acid
(vi) Haber process	F. Phosphoric acid
	G. Sulfuric acid

SECTION : B (54marks)

3. (a) During experiment, egg shells were dissolved in a beaker contained dilute hydrochloric acid and bubbles of a certain gas were evolved.

(vi) What a type of reaction took place in a beaker?
(vii) Write a balanced chemical equation for the reaction took place in a beaker
(viii) Write a net ionic equation for the reaction in (ii) above

(b) Use the information in the following scheme to answer the questions that follow



(i) Give condensed structure of compound C
(ii) Write a balance chemical equation for the **STEP 1**
(iii) What the type of reaction in **STEP 3**?
(iv) Suggest missing chemical reagent(s) for reaction in STEP 4

4. (a) A Doctor prescribe antacid tablets made from magnesium hydroxide to a student who was suffering from excess stomach acid:
(vi) Write a balanced chemical equation for the reaction took place in student's stomach after taking the tablets
(vii) If the normal acid content in the stomach is160milimolar and the stomach secrete 0.5 liters of gastric juice in a day, calculate the mass of tablets to be given to the student if the gastric juice had 220milimolar hydrochloric acid.
(viii) If each tablet of an antacid weighs 145mg of magnesium hydroxide, How many tablets will the student will be given
(ix) If a dose requires 1 tablet per every 12 hours, how many days should the students take to finish the full dose?

(b) Why air is said to be the mixture? (give three reasons)

5. (a) (i) Use electronic diagram to show the bon formation of carbon monoxide gas.
(ii) Find oxidation state of sulphur in sulphuric acid

(b) During the extraction of iron by using Hematite, the concentrated ore is strongly heated in excess air in shallow kilns and different chemical changes occur. With the chemical equations show that changes. (Give four points)

6. (a) "Both oxygen and hydrogen are neutral to the litmus paper. Instead of this similarity they have many differences". Verify this statement by giving three points.

(b) With the aid of chemical equations, briefly describe the following process.
(i) The removal of temporary hardness of water by boiling.
(ii) The removal of permanent hardness of water by addition of washing soda

7. (a)(i) souring of milk and melting of ice both are changes of substances. Based on the Properties of their changes distinguish them by giving four points
(ii) Why Non – luminous flame is formed when air hole of Bunsen burner is opened?
(iii) It is not possible to separate the mixtures of the liquids with equal boiling points,Why?

(b) Classify each of the following by state either are acid or base

(i) Lemon

(ii) Avocado

(iii) Apple

8. (a) Form three student wishes to monitor the rate of the following chemical reaction

$CuCO_3(s) + 2HCl(aq) \longrightarrow CuCl_2(aq) + CO_2(g) + H_2O(l)$

One of the properties that could be used to monitor the rate is the mass of $CuCO_3$ which decreases as reactants are converted to the products. Identify other four properties that could be used to monitor the rate of this reaction and describe the changes that would occur.

(b) MSHUA is a bus driver of NEWFORCE from Sumbawanga to Dar es salaam, when he started to drive from Sumbawanga Bus Terminal unfortunately crushed the MSALA whois a Boda boda driver at the gate of Bus Terminal and got injuries. MSHUA took someitems from a small box with a red cross and got out from the bus. He started to giveassistance to the MSALA before sent him to the Kristu Mfalme Hospital.

(i) What the action done by MSHUA after got out from the Bus?

(ii) Give four importance of action done by MSHUA to MSALA after got out from the Bus.

(iii) Identify any four items that were used by MSHUA when assist the Boda boda driver

SECTION : C (30marks)

9. "Fertile soil is not necessary to be productive". Justify this statement by giving six points.

10. Dar es salaam city has the problem of liquid, gaseous and solid wastes due to high population of the people who engaged in different economic and social activities that contribute it in production of such waste. You as Environmental Officer design six effective plans to be taken in order to solve such problem.

11. (a) (i) Cathode and Anode both are electrodes. During electrolysis electrode is one of main components of electrolytic cell. How can you contrast them by giving three points?

(ii) "Electrolysis of Sulphuric acid is refers to as electrolysis of water", by using Chemical equations for the reactions takes place during its electrolysis verify this statement.

(b) Give explanations for each of the following phenomenon

(i) The solid sodium chloride is a poor conductor of electricity while in a molten state, it readily conduct electricity

(ii) Distilled water does not conduct electricity while rain water does

(c) CHAKUBANGA is a chemical engineer who work at CHEMITECH PLANT that deal with production of hydrogen gas, He passed an electric current in a series through copper (II) sulphate and Sulphuric acid solutions. In both electrolytes, inert electrode was used.If $67200cm^3$ of hydrogen gas at STP were produced for several hours, what the volume of oxygen gas would also be produced?